Investigation Report

A-Mab Drug Substance — INV-001

Quality Assurance (QA)

2026-09-18

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Approvals

Table 2: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Quality Assurance (investigator)	—	—	—
Reviewer	Manufacturing Science & Technology	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Statistics	—	—	—
Approver	Head of Quality	—	—	—

Revision history

Version	Effective date	Description of change
1.0	2026-09-18	Initial issue

Abbreviations

AEX, anion exchange chromatography; BLA, Biologics License Application; CAPA, corrective and preventive action; CEX, cation exchange chromatography; Cpk, process capability index; CQA, critical quality attribute; DoE, design of experiments; HCP, host cell protein; HMW, high molecular weight species; LMW, low molecular weight species; MSAT, Manufacturing Science & Technology; MVM, minute virus of mice; NOR, normal operating range; nrCE-SDS, non-reduced capillary electrophoresis with sodium dodecyl sulfate; PAR, proven acceptable range; QA, Quality Assurance; SD, standard deviation; SEC, size-exclusion chromatography; SOP, standard operating procedure; VI, viral inactivation; XMuLV, xenotropic murine leukemia virus.

Summary

On 2026-08-03, Quality Assurance opened INV-001 during the review of PCMR-001 Version 1.0 for BLA readiness. The review found that the acceptance criteria against which each step was judged in the proven acceptable range analysis were not in any protocol approved before the studies ran. They had been set at analysis time, after the data were known. The investigation was classified as major.

Six findings are numbered F1 to F6 in §4. Four concern the criteria themselves. They were defined after the data were seen (F1), the safety factors that divided them differed by step and let every step pass at its own set-point (F2), the bioreactor limits were computed from the capability data they were then used to judge (F3), and acidic variants were assessed for capability against one end of a two-sided range (F4). F5 concerns the records, whose title blocks misstated the dates on which the plans were approved and which gave no execution dates. F6 concerns the method for fragments, where SEC read high in the capture matrices and the method decision AMD-001 was not carried into the documents.

The root cause is that PCMP-001 Version 1.0 required a criterion for each step but gave no rule for deriving one. The criteria were therefore derived during the analysis, by the people analysing the data (§5).

The investigation derived intermediate acceptance criteria a priori, carried back from the drug-substance criteria through the platform worst case of each downstream step, with no margin, and re-analysed the same models against them. The corrective actions reissue PCMP-001 at Version 2.0 with these criteria and reissue every other document in the package against them, with fragments measured by nrCE-SDS under AMD-001. They are due on 2026-09-30 and were open at closure. The preventive actions revise SOP-1002 so that a characterization protocol cannot be approved without its criteria (§7).

Of the 61 proven acceptable ranges that Version 1.0 reported with the other parameters varying within their NORs, 29 are wider in the re-analysis, 4 are narrower, 3 have moved and 25 are unchanged. In the re-analysis, the lowest capability index of the drug substance is that of acidic variants, at 1.03, against its lower limit. No GMP batch was dispositioned on these documents, and there is no product impact (§6).

1 Event description

PCMR-001 Version 1.0 and the reports it consolidates were approved on 2026-07-24. On 2026-08-03, QA reviewed PCMR-001 for use in the BLA. The review traced the in-process criteria in the control strategy of PCMR-001 back to the step plans and step reports. It found that none of the criteria could be traced to a plan as it was approved before execution, and that the master plan gave no rule from which the criteria could have been derived in advance. QA opened INV-001 the same day.

The 33 studies that Version 1.0 reported were executed between 2025-08-27 and 2026-05-15, and each of them closed before the Version 1.0 reports were approved.

Two immediate actions were taken on the day the investigation was opened.

1. The Version 1.0 documents were placed on hold for BLA use. They were kept unchanged, as the record under investigation.
2. No analysis that depended on the Version 1.0 step criteria was to be issued until the investigation had concluded.

The investigation was closed on 2026-09-18.

2 Scope

The investigation covers the twenty Version 1.0 documents of the A-Mab characterization package, listed in Table 2.1. Within them it covers four subjects. These are the step acceptance criteria used in the proven acceptable range analyses, with their derivation and their place in the protocols, the method used for fragments, the capability assessment of the drug substance, and the execution records of the studies.

Table 2.1: Documents in the scope of the investigation, all at Version 1.0 when it was opened.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
RA-001	Pre-Characterization Process Risk Assessment	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)

Document ID	Document class	Subject
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

Three things are outside the scope. The computation of a proven acceptable range itself, which is how the other parameters are held or varied while one parameter is scanned, is not reviewed here. The experiments are not reviewed either. Every study that Version 1.0 reported closed before the Version 1.0 reports were approved, so the re-analysis in §6 uses the same characterization data as Version 1.0. One further study ran during the investigation. The re-analysis of the Protein A scale-down qualification found that the 3 planned set-point runs could never demonstrate equivalence against the margin of PCP-005, because the confidence interval alone was wider than the margin (DEV-005-03). For this reason, 6 supplementary set-point runs (study S-PA-SDM2) were made from 2026-09-07 to 2026-09-11, and the margin was not changed. These runs add scale-down data only and change no characterization data that Version 1.0 analysed. They are a further action that the re-analysis needed, recorded in PCR-005, and not a corrective action of this investigation. Finally, the study deviations recorded in the step reports keep the disposition those reports give them.

3 Background

3.1 The master plan

PCMP-001 Version 1.0 required that each step be judged against a criterion that applies to it. It did not say how that criterion was to be obtained. Its §7 left the threshold to the step plans.

> Where a threshold depends on the step, on the attribute or on the precision of the method, the criterion is defined in the plan of that step and the reason is given there.

The same section stated that the criterion for a response is the in-process limit of the step where one has been established, and the drug-substance criterion where none has. It gave neither a rule nor a value for an in-process limit.

3.2 The Protein A analysis

The need for step criteria became apparent during the analysis of the Protein A study. The capture pool carries host cell protein at a level far above the drug-substance criterion of 100 ng/mg. Protein A capture removes most of the host cell protein of the harvest, and part of the remainder is carried into the eluate bound to the antibody. That level is acceptable in an intermediate, because cation exchange and anion exchange remove most of it afterwards. Judged against the drug-substance criterion, no part of the Protein A operating region would have been acceptable. PCMR-001 Version 1.0 put the argument in one sentence.

> The drug substance limit is the wrong yardstick for an intermediate, and the campaign uses a criterion for each step that applies to that step.

What the investigation examined is how the step criteria were then set, and when.

3.3 The Version 1.0 criteria

Version 1.0 set 15 step criteria at 6 steps, by four rules. Table 3.1 gives each criterion with its rule, beside the Version 2.0 criterion that replaced it. The Version 2.0 columns are discussed in §6.

1. For an impurity that downstream steps clear, the drug-substance criterion was carried back through the clearance the downstream steps delivered at their set-points in this campaign, and the result was divided by a safety factor. This rule (R1) set 6 criteria.
2. For an attribute that no downstream step changes, the criterion was an alert limit at the capability mean plus 2.5 SD. This rule (R2) set 4 criteria, all at the bioreactor.
3. For viral clearance at low-pH viral inactivation and anion exchange, the criterion was the cumulative requirement less the nominal clearance of the other credited steps. This rule (R3) set 3 criteria.
4. For viral clearance at virus filtration, the criterion was the nominal clearance claimed for the step less 0.35 log10 for assay variability. This rule (R4) set 2 criteria.

R1, R3 and R4 use the nominal results of this campaign, and R2 uses its capability data. Every Version 1.0 criterion therefore depends on data from the studies it was used to judge.

Table 3.1: Acceptance criteria for the step outputs in Version 1.0 and Version 2.0.

Step	Attribute	Version 1.0[b]		Version 2.0	
		Criterion	Rule	Criterion	Basis
Production bioreactor	Acidic variants (%)	18.0–25.8	R2	18.0–39.0	back-calculated
	Afucosylation (%)	2.00–9.43	R2	2.00–13.0	drug substance
	Aggregates, HMW (%)	≤ 1.86	R1 margin 6.8	≤ 6.83	back-calculated, tightened on verification
	Galactosylation (%)	10.0–30.1	R2	10.0–40.0	drug substance
	High mannose (%)	3.00–7.28	R2	3.00–10.0	drug substance
Protein A	Leached Protein A (ppm)	none[a]		≤ 158	back-calculated
	Fragments, LMW (%)	not assessed[c]		≤ 2.45	back-calculated
	Host cell protein (ng/mg)	≤ 10,724	R1 margin 4.5	≤ 16,000	back-calculated
Low-pH viral inactivation	Acidic variants (%)	none[a]		18.0–40.0	drug substance
	Aggregates, HMW (%)	≤ 2.17	R1 margin 6.0	≤ 7.53	back-calculated, tightened on verification
	Fragments, LMW (%)	not assessed[c]		≤ 2.60	back-calculated
	XMuLV clearance (log10)	≥ 4.93	R3	≥ 6.60	platform claim
Cation exchange	Aggregates, HMW (%)	≤ 1.00	R1 margin 5.0	≤ 5.00	drug substance
	Host cell protein (ng/mg)	≤ 193	R1 margin 3.2	≤ 400	back-calculated
	Fragments, LMW (%)	not assessed[c]		≤ 2.00	drug substance
Anion exchange	Host cell protein (ng/mg)	≤ 21.7	R1 margin 4.6	≤ 100	drug substance
	Fragments, LMW (%)	not assessed[c]		≤ 2.00	drug substance
	MVM clearance (log10)	≥ 3.28	R3	≥ 4.00	platform claim
	XMuLV clearance (log10)	≥ 3.78	R3	≥ 5.50	platform claim
Virus filtration	MVM clearance (log10)	≥ 4.97	R4	≥ 4.62	platform claim
	XMuLV clearance (log10)	≥ 5.47	R4	≥ 4.62	platform claim

[a] Version 1.0 set no step criterion and judged the step against the drug-substance criterion.
[b] Applied in the Version 1.0 reports at analysis. No approved Version 1.0 plan contained these criteria (INV-001 finding 1).
[c] Version 1.0 did not report fragments. The attribute was added at Version 2.0.
R1 Version 1.0 rule: nominal back-calculation / margin.
R2 Version 1.0 rule: mean + 2.5 sd of capability.
R3 Version 1.0 rule: requirement - nominal of other steps.
R4 Version 1.0 rule: nominal claim less 0.35 log10 for assay variability.

4 Investigation

4.1 Approach

The sequence in which the criteria were set was reconstructed from the analysis records, from the study chronology and from the authors of the Version 1.0 reports. Each criterion in Table 3.1 was recomputed from its stated rule and compared with the value printed in the documents. The capability assessment was repeated against the drug-substance criteria as the register prints them. The execution windows of the studies were compared with the calibration records of the equipment they used. The method record for fragments was reviewed against the method decision AMD-001 and the comparability study S-CEX-FRC. Every recomputed criterion agreed with the printed value.

4.2 F1. Criteria defined after the data were seen

The step criteria were defined after the data were seen and outside the approved protocols.

Each step plan was approved before its studies began, as Table 4.3 in §4.6 shows. The plan for Protein A, for example, was approved on 2025-11-03, and its first study started on 2025-11-05. None of the step criteria was in a plan as approved. The criteria were introduced later, while the Protein A data were being analysed, when it became clear that the capture pool could not be judged against the drug-substance criterion (§3.2). They were then written into the text of every step plan when the Version 1.0 package was compiled, and the approved plans were not revised to carry them. The Version 1.0 plans therefore read as if the criteria had been approved before execution, which they had not.

The master plan, approved on 2025-08-04, did not prevent this. As §3.1 shows, it required a step criterion without fixing one. The record therefore holds no protocol, approved before execution, that contains a step criterion. A criterion added to a protocol after its data are known does not meet the purpose of a protocol (U.S. Food and Drug Administration 2011).

4.3 F2. Safety factors chosen so that each step passed at its set-point

Each criterion set by R1 was the drug-substance criterion carried back through the nominal downstream clearance of this campaign, then divided by a safety factor. The factors ranged from 3.2 to 6.8, and no document records why they differ.

Table 4.1 sets each factor against the value the step's own response surface model predicts at its set-point. The last column gives the largest factor at which that set-point value would still have met the criterion. Every step output meets its criterion at its own set-point. Had the largest factor used, 6.8, been applied to every step, the set-point value would have failed its criterion for 4 of the outputs in Table 4.1. The analysis records show that a factor was reduced where the limit would otherwise have fallen below the value the step produces at its set-point. The factors were therefore chosen with the result in view.

Table 4.1: Safety factors of the Version 1.0 step criteria against the value each step produces at its set-point. The third column is the concentration above which the downstream steps, at the nominal clearance of this campaign, can no longer bring the batch to specification.

Step	Attribute	Highest level downstream steps bring to specification	Safety factor	Version 1.0 criterion	Predicted at set-point	Largest factor met at set-point
Production bioreactor	Aggregates, HMW (%)	12.7	6.8	≤ 1.86	1.52	8.32
Protein A	Host cell protein (ng/mg)	48,260	4.5	≤ 10,724	7,216	6.69
Low-pH viral inactivation	Aggregates, HMW (%)	13.0	6.0	≤ 2.17	1.91	6.81
Cation exchange	Aggregates, HMW (%)	5.00	5.0	≤ 1.00	0.75	6.66
Cation exchange	Host cell protein (ng/mg)	619	3.2	≤ 193	141	4.38
Anion exchange	Host cell protein (ng/mg)	100.0	4.6	≤ 21.7	14.8	6.75

That level itself rests on nominal clearance. A criterion carried back through the clearance each downstream step delivers at its set-point assumes that every downstream step performs at its set-point. The safety factor stood in for the worst case

of downstream performance without being derived from it. The viral clearance criteria of R3 have the same weakness. Each was the cumulative requirement less the nominal clearance of the other steps in this campaign.

4.4 F3. Capability limits judged against the data they came from

The four bioreactor criteria set by R2 were the mean plus 2.5 SD of the commercial-scale capability simulation. Table 4.2 recomputes each one. The recomputed value and the printed criterion agree to the second decimal place.

Table 4.2: Version 1.0 bioreactor criteria recomputed from the capability data. Mean and SD are those of the simulated drug substance at commercial scale.

Attribute	Capability mean	Capability SD	Mean + 2.5 SD	Version 1.0 upper criterion	Drug-substance criterion
Acidic variants (%)	22.30	1.386	25.76	25.76	18.0–40.0
Afucosylation (%)	7.98	0.580	9.43	9.43	2.00–13.0
Galactosylation (%)	25.98	1.631	30.06	30.06	10.0–40.0
High mannose (%)	5.99	0.515	7.28	7.28	3.00–10.0

The capability simulation draws every parameter inside its NOR. The proven acceptable range analysis with the other parameters in their NORs then asked whether the predictive bound stayed within the alert limit while the other parameters varied inside the same NOR. For a normal distribution, about 0.62 % of the batches lie above a one-sided limit at the mean plus 2.5 SD, whatever the process. A limit of this kind is an alert limit for trending. It is not an acceptance criterion, because it carries no information about what the product needs.

The consequence is visible in PCR-003 Version 1.0. It reported a corner of the NOR at which predicted afucosylation exceeded its alert limit while meeting the drug-substance criterion by a wide margin. Several proven acceptable ranges at the bioreactor did not cover the whole NOR for the same reason (§6.3).

4.5 F4. Acidic variants assessed against one end of a two-sided range

The register gives acidic variants a two-sided criterion of 18 % to 40 %, and the Version 1.0 capability table printed that range. The capability index beside it was

computed against the upper end alone and printed as 4.3. The simulated drug substance has a mean of 22.3 % and an SD of 1.39 %, so the lower end of the range is much nearer the distribution than the upper end. Computed against both ends, the index is 1.03, and the lower end governs it.

Two statements in Version 1.0 rested on the one-sided value. PCMR-001 reported the lowest capability index of the drug substance as 1.51, for MVM clearance. It also reported that no simulated batch fell outside an acceptance criterion. Against both ends of the acidic variant range, the simulation puts one batch in 2,000 outside it.

4.6 F5. Approval and execution dates missing from the records

The Version 1.0 records misstated when the plans were approved and did not state when the studies ran. Every Version 1.0 title block carried the date 2026-07-24. That was the approval date of the reports and the date on which the package was compiled. For the plans, it replaced the date of approval. Table 4.3 sets the approval date of each plan against the start of its first study and against the date its title block printed. Every plan was approved before its first study began. Read from the title blocks, every plan appeared to have been approved after its studies had run.

Table 4.3: Approval date of each step plan, against the start of its first study and the date printed in its Version 1.0 title block.

Plan	Step	Approved	First study started	Date in Version 1.0 title block
PCP-003	Production bioreactor	2025-08-25	2025-08-27	2026-07-24
PCP-004	Harvest	2025-10-20	2025-11-03	2026-07-24
PCP-005	Protein A	2025-11-03	2025-11-05	2026-07-24
PCP-006	Low-pH viral inactivation	2026-01-05	2026-01-07	2026-07-24
PCP-007	Cation exchange	2026-02-02	2026-02-04	2026-07-24
PCP-008	Anion exchange	2026-02-02	2026-02-04	2026-07-24
PCP-009	Virus filtration	2026-04-13	2026-04-15	2026-07-24
PCP-010	UF/DF	2026-02-23	2026-02-25	2026-07-24

The Version 1.0 reports gave no execution date for any study. Their equipment tables recorded each system as in calibration at execution and gave the next calibration due date. Every one of these due dates precedes the date in the title blocks. A reader of the record could not confirm calibration at the time of use, and the tables appeared to cite expired calibrations.

The execution windows were recovered from the study records and are given in Table 4.4. Each study in Table 4.4 closed before the calibration of the equipment it used fell due. The narrowest margin is at the UF/DF study, which closed on 2026-03-27

against a calibration due on 2026-04-22. No plan was approved after its studies, and none of these studies was run on equipment out of calibration.

Table 4.4: Execution windows of the studies that used instrumented systems, against the calibration due date and the date in the Version 1.0 title blocks.

Equipment	Study	Execution start	Execution end	Next calibration due	Date in Version 1.0 title block
EQ-CHR-118	S-CEX-RSM	2026-03-02	2026-03-27	2026-05-30	2026-07-24
EQ-BRX-205	S-BRX-SDM	2025-08-27	2025-09-05	2026-06-14	2026-07-24
EQ-BRX-205	S-BRX-UNI	2025-10-06	2025-10-17	2026-06-14	2026-07-24
EQ-BRX-205	S-BRX-SCR	2025-09-08	2025-10-03	2026-06-14	2026-07-24
EQ-BRX-205	S-BRX-RSM	2025-10-20	2025-12-12	2026-06-14	2026-07-24
EQ-TFF-142	S-UFDF-SDM	2026-02-25	2026-03-06	2026-04-22	2026-07-24
EQ-TFF-142	S-UFDF-UNI	2026-03-09	2026-03-27	2026-04-22	2026-07-24

4.7 F6. Fragments by SEC in the capture matrices

In the comparability study S-CEX-FRC at cation exchange, executed from 2026-03-02 to 2026-03-13, fragments were measured by SEC and by nrCE-SDS on the same samples. Table 4.5 gives every replicate.

On average, SEC read fragments higher than nrCE-SDS by 0.4 to 0.5 percentage points in the Protein A pool, the VI pool and the CEX load. From the CEX pool onward, the mean difference between the two methods was no more than 0.1 percentage points. The difference follows the host cell protein content of the sample. Part of the host cell protein elutes in the low molecular weight window of SEC, where it is counted with the clipped antibody species. The excess can therefore be no larger than the host cell protein in the sample. Cation exchange removes most of the host cell protein, and from its pool onward the excess is close to zero. In the Protein A pool, SEC reported a mean of 1.3 % fragments, where nrCE-SDS reported 0.9 %. The excess there is about 47 % of the fragment level that nrCE-SDS measures. SEC fragment results from the capture matrices are therefore biased high by an amount that matters for a fragment criterion.

The method decision AMD-001 was taken on 2026-03-20 on these data. From that date, fragments were measured by nrCE-SDS in every study, at every step, so that

Table 4.5: Fragments by SEC and by non-reduced CE-SDS on the same samples (study S-CEX-FRC).

Sample	Replicate	Fragments, LMW (%)		SEC minus nrCE-SDS
		SEC	nrCE-SDS	
Protein A pool	1	1.32	0.90	+0.42
	2	1.42	1.10	+0.32
	3	1.32	0.82	+0.50
	4	1.29	0.83	+0.46
VI pool	1	1.43	0.84	+0.59
	2	1.39	0.88	+0.51
	3	1.42	0.88	+0.54
	4	1.49	1.02	+0.47
CEX load	1	1.36	0.85	+0.52
	2	1.52	1.03	+0.49
CEX fraction F2	1	1.65	1.56	+0.09
CEX fraction F3	1	0.64	0.76	-0.11
CEX fraction F4	1	0.40	0.49	-0.09
CEX fraction F5	1	0.42	0.32	+0.10
CEX pool	1	0.61	0.62	-0.01
	2	0.70	0.57	+0.13
	3	0.51	0.61	-0.10
	4	0.70	0.72	-0.02
AEX pool	1	0.55	0.65	-0.10
	2	0.68	0.70	-0.02

nrCE-SDS reports the sum of clipped species. The partial molecules seen under SDS (HHL, HH, HL, free light chain), which come from antibody lacking one or more interchain disulfide bonds, are excluded. This makes the two methods comparable.

Table 4.6: Methods used for size variants, by step and study.

Step	Attribute	Method used in study	
		Screening	Response surface
Production bioreactor	Aggregates, HMW	SEC	
	Fragments, LMW	not in scope[f)]	
Harvest	Aggregates, HMW	no DoE[e)]	
	Fragments, LMW	no DoE[e)]	
Protein A	Aggregates, HMW	SEC	
	Fragments, LMW	SEC[a)]	SEC[a)], nrCE-SDS[b)]
Low-pH viral inactivation	Aggregates, HMW	SEC	
	Fragments, LMW	SEC[a)]	SEC[a)], nrCE-SDS[b)]
Cation exchange	Aggregates, HMW	SEC	
	Fragments, LMW	SEC	SEC, nrCE-SDS[c)]
Anion exchange	Aggregates, HMW	SEC	
	Fragments, LMW	nrCE-SDS[d)]	nrCE-SDS[d)]
Virus filtration	Aggregates, HMW	SEC	
	Fragments, LMW	nrCE-SDS	nrCE-SDS
UF/DF	Aggregates, HMW	no DoE[e)]	
	Fragments, LMW	no DoE[e)]	

[a)] In this matrix part of the host cell protein elutes in the SEC low molecular weight window. Reported for information only (AMD-001).
[b)] Retained response-surface samples, tested 2026-04-06 to 2026-04-17.
[c)] Both methods on the same samples, the comparability study S-CEX-FRC.
[d)] Re-executed design (DEV-008-01). The invalidated first execution measured fragments by SEC.
[e)] Univariate characterization only.
[f)] Fragments were not an attribute of this study.

one method spans the train. The CEX response surface study, which ran from 2026-03-02 to 2026-03-27 and so spanned the decision, measured fragments by both methods in parallel. The retained response surface samples of Protein A and viral inactivation were retested by nrCE-SDS from 2026-04-06 to 2026-04-17. The screening samples had not been retained. Their SEC results are kept and marked as not used. The re-executed anion exchange design (DEV-008-01) started on 2026-03-23, after the decision, and measured fragments by nrCE-SDS. Its invalidated first execution had used SEC. Fragments were not in the scope of the bioreactor design. Table 4.6 shows which method ran in which study.

The decision and the retests predate Version 1.0. However, the Version 1.0 documents did not report fragments, their register carried no fragment attribute, and none of them cited AMD-001. No procedure required that a method decision taken during a campaign be assessed against the approved plans and, where it changed them, be put into effect by a plan amendment. AMD-001 therefore never reached the documents.

5 Root cause

The root cause is that PCMP-001 Version 1.0 required a criterion for each step but gave no rule for deriving one (F1). When the Protein A analysis showed that an intermediate cannot be judged against a drug-substance criterion, the criteria were derived during the analysis, by the people analysing the data, with the results in view. Every rule they chose drew on data from the studies being judged (§3.3).

Four causes contributed.

1. SOP-1002 did not require that the acceptance criteria and their basis be approved in the protocol before execution. Criteria set after execution could therefore be added to the plan text when the package was compiled, without a revision of the approved plan (F1).
2. Neither the plans nor the master plan defined the downstream performance that a step criterion protects against. A safety factor was used in place of a worst case, and its size was set step by step (F2). Capability data were used where no downstream step changes the attribute (F3).
3. No second review checked the specification type used in the capability calculation against the register (F4), or the dates in the title blocks and study records against the approval and equipment records (F5).
4. No procedure required that a method decision be assessed against the approved plans and put into effect by plan amendment (F6).

6 Impact assessment

6.1 The Version 2.0 criteria

The intermediate acceptance criteria were derived a priori by the investigation, by one rule, and CA1 issues them in PCMP-001 Version 2.0. Each drug-substance criterion is carried backward through the train, using the worst case each downstream step showed on the platform (X-Mab, Y-Mab and Z-Mab). That is the lowest fold clearance and the largest increase. No margin is applied. The glycans keep their drug-substance criterion, because no downstream step changes them. Each viral clearance step is held to its platform claim, and the claims sum to the cumulative requirement, 16.72 against 16.7 log10 for XMuLV and 8.62 against 8.6 log10 for MVM.

Table 6.1 gives the 21 criteria with their derivation. Of these, 8 are the drug-substance criterion applied directly, 8 are carried back through the platform worst case, and 5 are platform viral clearance claims.

Applying these criteria to studies already executed is acceptable because their derivation uses no data from those studies. The drug-substance criteria come from the register and the worst cases from the platform products.

6.2 Verification of the platform worst cases

The platform worst cases are assumptions about A-Mab, and each was checked against the data of the step it describes. The test is the value the response surface model predicts within the NOR. The value observed at the edges of the characterization range is shown for information. For host cell protein at cation exchange, the edge value of 30.3 falls short of the platform fold clearance of 40.0, but the value predicted within the NOR, 43.6, meets it. Table 6.2 gives the result. Of the 7 assumptions that could be tested, 6 were confirmed.

One assumption was not confirmed. The platform worst case for aggregate at cation exchange is a fold clearance of 2.0. Within its NOR, the response surface model of this campaign predicts a fold clearance of 1.51. The criteria upstream of cation exchange were recalculated with the A-Mab value, and 2 aggregate criteria were tightened, at the production bioreactor and low-pH viral inactivation steps. Whether any proven acceptable range moved because of this is stated in the affected reports, from their own analysis.

The criterion for leached Protein A at the Protein A step rests on two platform worst cases, for cation exchange and anion exchange, that no response of this campaign

Table 6.1: Intermediate acceptance criteria and their derivation from the drug-substance criteria.

Attribute	Drug-substance criterion	Step output	Downstream worst case		Intermediate acceptance criterion	Basis
			Clearance fold	Increase		
Host cell protein (ng/mg)	≤ 100	Protein A	160	0.00	≤ 16,000	Back-calculated
		Cation exchange	4.00	0.00	≤ 400	Back-calculated
		Anion exchange	1.00	0.00	≤ 100	Drug substance[a]
Aggregates, HMW (%)	≤ 5.00	Production bioreactor	1.51	0.70	≤ 6.83	Back-calculated, tightened[b]
		Low-pH viral inactivation	1.51	0.00	≤ 7.53	Back-calculated, tightened[b]
		Cation exchange	1.00	0.00	≤ 5.00	Drug substance[a]
Fragments, LMW (%)	≤ 2.00	Protein A	1.30	0.15	≤ 2.45	Back-calculated
		Low-pH viral inactivation	1.30	0.00	≤ 2.60	Back-calculated
		Cation exchange	1.00	0.00	≤ 2.00	Drug substance[a]
		Anion exchange	1.00	0.00	≤ 2.00	Drug substance[a]
Leached Protein A (ppm)	≤ 5.00	Protein A	31.6	0.00	≤ 158	Back-calculated
Acidic variants (%)	18.0–40.0	Production bioreactor	1.00	1.00	18.0–39.0	Back-calculated
		Low-pH viral inactivation	1.00	0.00	18.0–40.0	Drug substance[a]
Afucosylation (%)	2.00–13.0	Production bioreactor	1.00	0.00	2.00–13.0	Drug substance[a]
Galactosylation (%)	10.0–40.0	Production bioreactor	1.00	0.00	10.0–40.0	Drug substance[a]
High mannose (%)	3.00–10.0	Production bioreactor	1.00	0.00	3.00–10.0	Drug substance[a]
XMuLV clearance (log10)	≥ 16.7[d]	Low-pH viral inactivation	not applicable		≥ 6.60	Platform claim[c]
		Anion exchange	not applicable		≥ 5.50	Platform claim[c]
		Virus filtration	not applicable		≥ 4.62	Platform claim[c]
MVM clearance (log10)	≥ 8.60[d]	Anion exchange	not applicable		≥ 4.00	Platform claim[c]
		Virus filtration	not applicable		≥ 4.62	Platform claim[c]

[a] No downstream step changes the attribute, so the drug-substance criterion applies.
[b] The response-surface model of a downstream step predicted, within its NOR, a worse value than the platform worst case. The A-Mab value replaced it.
[c] Log reduction claimed for this step in the modular viral clearance claim.
[d] Cumulative requirement across the steps that clear the virus.

Table 6.2: Platform worst cases assumed by the intermediate acceptance criteria, against this campaign's results.

Step	Attribute	Worst case	Platform	This campaign		Confirmed
				At the edges	Within the NOR	
Low-pH viral inactivation	Aggregates, HMW (%)	Increase, largest	0.70	0.84	0.52	Yes
	Fragments, LMW (%)	Increase, largest	0.15	0.22	0.10	Yes
	Acidic variants (%)	Increase, largest	1.00	0.60	0.40	Yes
Cation exchange	Host cell protein (ng/mg)	Clearance fold, lowest	40.0	30.3	43.6	Yes
	Aggregates, HMW (%)	Clearance fold, lowest	2.00	1.01	1.51	No[b]
	Fragments, LMW (%)	Clearance fold, lowest	1.30	1.28	1.42	Yes
	Leached Protein A (ppm)	Clearance fold, lowest	3.16	n.t.[a]	n.t.[a]	n.t.[a]
Anion exchange	Host cell protein (ng/mg)	Clearance fold, lowest	4.00	3.63	5.26	Yes
	Leached Protein A (ppm)	Clearance fold, lowest	10.0	n.t.[a]	n.t.[a]	n.t.[a]

[a] Not a response of the response-surface study at this step.
[b] The criteria upstream of this step were recalculated with the value predicted within the NOR.

could test (n.t. in Table 6.2). This is acceptable. The model predicts 2.87 ppm at the set-point and at most 3.22 ppm over the NOR (Table 6.4). That is below the drug-substance criterion of 5 ppm itself, so the pool would meet the drug-substance criterion even if no downstream step cleared any leached Protein A.

6.3 Impact by document

All twenty documents in Table 2.1 are to be reissued at Version 2.0, effective 2026-09-30 (CA1 to CA3). PCMP-001 will define the criteria of §6.1. The step plans PCP-003 to PCP-010 will state them with their basis. The step reports will judge the same models against them, and PCMR-001 will consolidate the result. PTP-001 and RA-001 are reissued with the package. Each Version 2.0 document will say what changed in it under “Changes in this revision” and refer to the findings by number.

Table 6.3 compares the proven acceptable ranges, with the other parameters varying within their NORs, in each report with a DoE, as Version 1.0 reported them and as the re-analysis gives them. The reports without a DoE, PCR-004 and PCR-010, contain no such ranges. Their revision will record the execution windows (F5).

Table 6.3: Proven acceptable ranges with the other parameters varying within their NORs, in Version 1.0 and in the re-analysis, by report. New entries are the fragment ranges measured by nrCE-SDS.

Report	Step	Entries in Version 1.0	Unchanged	Wider	Narrower	Moved	New in Version 2.0
PCR-003	Production bioreactor	20	3	13	1	3	0
PCR-005	Protein A	8	3	5	0	0	4
PCR-006	Low-pH viral inactivation	9	6	1	2	0	3
PCR-007	Cation exchange	8	2	6	0	0	4
PCR-008	Anion exchange	12	9	2	1	0	4
PCR-009	Virus filtration	4	2	2	0	0	2

Where a range changed, it widened far more often than it narrowed. Of the 29 wider ranges, 13 are at the bioreactor. There, 11 glycan ranges widened because the R2 capability limits (F3) gave way to the drug-substance criteria, and 2 aggregate ranges widened because an R1 limit with a safety factor (F2) gave way to a back-calculated criterion. Most of the wider ranges at the other steps follow from the R1 safety factors.

At cation exchange, the Version 1.0 criterion of 1.00 % HMW was the drug-substance criterion divided by a safety factor of 5.0. Against it, the proven acceptable range for protein load, with the other parameters varying within their NORs, was 10–27.6 g/L resin, which did not reach the upper end of the protein load NOR at 32 g/L resin. Against the Version 2.0 criterion of 5 %, the range is 10–40 g/L resin. The upper load NOR is supported. At the bioreactor, the range for dissolved CO2 judged on galactosylation, computed the same way, was 74.5–160 mmHg at Version 1.0, which did not reach the lower end of the NOR at 70 mmHg. Judged against the drug-substance criterion, it is 40–160 mmHg.

Of the 4 ranges that narrowed, 2 at low-pH viral inactivation and 1 at anion exchange narrowed because a viral clearance criterion rose to its platform claim, and 1 at the bioreactor narrowed although no criterion there became tighter, because acidic variants are now judged against both ends of their range (F4). At low-pH viral inactivation the XMuLV criterion rose from 4.93 to the platform claim of 6.6 log10, and the inactivation pH range narrowed from 3.2–3.96 to 3.2–3.59 as a result. At the bioreactor no criterion became tighter, yet 4 of the acidic variant ranges changed and none widened. Acidic variants are now judged against both ends of their range (F4), and the lower limit of 18 % constrains them.

Table 6.4 gives each re-analysed step output against its Version 2.0 criterion, from the response surface model over the NOR. Of the 21 outputs, 19 meet their criterion across the NOR. The exceptions are acidic variants at the production bioreactor step (predicted down to 17.7 at a corner of the NOR) and XMuLV clearance at the low-pH viral inactivation step (predicted down to 6.43 at a corner of the NOR). Each is met at the set-point, and the affected report states the consequence for its ranges. Table 6.5 sets the viral clearance claims against the step results of this campaign, and every claim is met.

Table 6.4: Re-analysed step outputs against their Version 2.0 criteria. The range is that predicted by the response surface model over the corners and centre of the NOR.

Step	Attribute	Version 2.0 criterion	At set-point	Range over the NOR	Met across the NOR
Production bioreactor	Afucosylation (%)	2.00–13.0	8.04	6.40 to 9.43	Met
Production bioreactor	Galactosylation (%)	10.0–40.0	25.8	21.5 to 31.8	Met
Production bioreactor	High mannose (%)	3.00–10.0	5.89	5.19 to 6.90	Met
Production bioreactor	Acidic variants (%)	18.0–39.0	22.0	17.7 to 25.7	Not met
Production bioreactor	Aggregates, HMW (%)	≤ 6.83	1.52	1.20 to 1.84	Met
Protein A	Host cell protein (ng/mg)	≤ 16,000	7,216	3,550 to 13,553	Met

Step	Attribute	Version 2.0 criterion	At set-point	Range over the NOR	Met across the NOR
Protein A	Leached Protein A (ppm)	≤ 158	2.87	2.46 to 3.22	Met
Protein A	Fragments, LMW (%)	≤ 2.45	0.91	0.86 to 0.97	Met
Low-pH viral inactivation	Aggregates, HMW (%)	≤ 7.53	1.91	1.76 to 2.08	Met
Low-pH viral inactivation	XMuLV clearance (log10)	≥ 6.60	6.87	6.43 to 7.29	Not met
Low-pH viral inactivation	Acidic variants (%)	18.0–40.0	20.6	20.5 to 20.7	Met
Low-pH viral inactivation	Fragments, LMW (%)	≤ 2.60	0.98	0.97 to 1.01	Met
Cation exchange	Aggregates, HMW (%)	≤ 5.00	0.75	0.53 to 1.27	Met
Cation exchange	Host cell protein (ng/mg)	≤ 400	141	95.3 to 209	Met
Cation exchange	Fragments, LMW (%)	≤ 2.00	0.62	0.56 to 0.69	Met
Anion exchange	Host cell protein (ng/mg)	≤ 100	14.8	9.26 to 22.2	Met
Anion exchange	XMuLV clearance (log10)	≥ 5.50	6.58	5.83 to 7.48	Met
Anion exchange	MVM clearance (log10)	≥ 4.00	4.87	4.20 to 5.65	Met
Anion exchange	Fragments, LMW (%)	≤ 2.00	0.61	0.58 to 0.65	Met
Virus filtration	MVM clearance (log10)	≥ 4.62	5.55	5.16 to 6.52	Met
Virus filtration	XMuLV clearance (log10)	≥ 4.62	6.10	6.04 to 6.27	Met

Table 6.5: Viral clearance claims against the step results of this campaign.

Step	Virus	Platform claim (log10)	Step result (log10)	Claim
Low-pH viral inactivation	XMuLV	≥ 6.6	7.10	Met
Anion exchange	XMuLV	≥ 5.5	5.95	Met
Anion exchange	MVM	≥ 4	4.71	Met
Virus filtration	MVM	≥ 4.62	5.32	Met
Virus filtration	XMuLV	≥ 4.62	5.82	Met

6.4 Capability

In the re-analysis, the capability of acidic variants was assessed against both ends of its range (F4). The lowest capability index of the drug substance moves from 1.51 for MVM clearance to 1.03 for acidic variants, against the lower limit of 18 %. PCMP-001 Version 2.0 sets a capability target of 1.33, which is common industry practice for a capable process, and a minimum of 1.00, the point at which a spread of three standard deviations reaches the criterion. Only acidic variants falls below the target, and every attribute meets the minimum. The re-analysis also gives the probability of a drug substance batch outside specification for each attribute, from the same simulation. For acidic variants the simulation puts one batch in 2,000 outside the range, and a normal fit gives 971 per million. No simulated batch falls outside the range of any other attribute.

6.5 Fragments

In the re-analysis, every study in scope for fragments is judged on nrCE-SDS, against the fragment criteria in Table 6.1. SEC fragment results from the Protein A and viral inactivation matrices will be reported for information only and not used to judge a step. They could not have supported a fragment criterion at those steps, because they include host cell protein (F6).

6.6 Product impact

No GMP batch of A-Mab was manufactured or dispositioned on the basis of these documents, and Version 1.0 was not used in a regulatory submission. The findings concern how the characterization data were judged. The experiments, the process and the drug-substance criteria that existed in Version 1.0 are unchanged. Version 2.0 adds one drug-substance criterion, for fragments, at ≤ 2.00 %. There is no product impact.

7 Corrective and preventive actions

The actions in Table 7.1 follow from the root cause and the contributing causes in §5. All six actions were open at closure and are tracked to their due dates in the CAPA system (International Council for Harmonisation 2008). The retest of retained samples by nrCE-SDS, which finished on 2026-04-17 under AMD-001, is not an action of this investigation. It is a prior action that F6 relies on.

Table 7.1: Corrective (CA) and preventive (PA) actions.

No.	Action	Owner function	Due	Status
CA1	Reissue PCMP-001 with a priori intermediate acceptance criteria	MSAT	2026-09-30	open at closure
CA2	Amend PCP-003 to PCP-010 with the criteria and the fragment method	MSAT	2026-09-30	open at closure
CA3	Reissue PCR-003 to PCR-010, PTP-001, RA-001 and PCMR-001 at Version 2.0	MSAT	2026-09-30	open at closure
PA1	Revise SOP-1002 so acceptance criteria are approved in the protocol before execution	Quality Assurance	2026-11-30	open at closure
PA2	Add to the report review checklist of SOP-4001 a check that every acceptance criterion is traceable to the approved protocol	Quality Assurance	2026-11-30	open at closure
PA3	Record execution dates and equipment calibration status in every study record	MSAT	2026-10-31	open at closure

CA1 removes the root cause for this campaign, and it also answers contributing cause 2. The rule in PCMP-001 Version 2.0 fixes every step criterion before any study is judged against it, defines the downstream worst case each criterion protects against, and uses no data from the studies it judges. CA2 amends the step plans with the criteria and the fragment method, which brings AMD-001 into the plans (F6). CA3 reissues the reports, which judge the same models against the new criteria and assess acidic variants against both ends of their range (F4).

PA1 removes the root cause, and contributing cause 1, for future campaigns. PA2 checks at report review that every acceptance criterion can be traced to the approved

protocol, which would have detected the criteria added after approval in F1. PA3 records execution dates and calibration status in every study record, which answers the dates part of contributing cause 3 (F5).

Two causes have no preventive action. No action adds a second review of the specification type used in a capability calculation, so F4 is corrected by CA3 for this campaign only. For this product, the two-sided assessment is fixed in PCMP-001 Version 2.0, so the error cannot recur in this campaign. No action creates the procedural step named in contributing cause 4 either. CA2 brings AMD-001 into the plans, but a future method decision would again depend on the people involved to reach the documents.

8 Effectiveness check

The corrective actions are verified effective when the Version 2.0 package is effective and QA has repeated the review that triggered this investigation, on PCMR-001 Version 2.0. That review checks three things. Every step criterion used in a proven acceptable range analysis must appear in PCMP-001 Version 2.0 with its basis, and no criterion may draw on data from the studies it judges. Every fragment result used to judge a step must be by nrCE-SDS. Every study must carry its execution window.

QA verifies the preventive actions by 2027-03-31. The check requires the next two characterization protocols approved with acceptance criteria and provenance, before execution. QA owns both checks.

9 Conclusion and closure

The investigation confirmed the classification as major. The criteria used to support the operating ranges in a BLA were not defined before the data were seen, and the analyses that rested on them could not be relied on as approved. The investigation found no product impact (§6.6).

The corrective actions are due with Version 2.0 on 2026-09-30, and the preventive actions by 2026-11-30. INV-001 was closed on 2026-09-18, with its remaining actions open and tracked to completion under the effectiveness check in §8.

References

International Council for Harmonisation. 2008. “ICH Q10: Pharmaceutical Quality System.” ICH.

U.S. Food and Drug Administration. 2011. “Guidance for Industry: Process Validation — General Principles and Practices.” FDA.